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A Study Comparing Machine Learning Models for Network Intrusion Detection

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Abstract: **Given the increasing volume and sophistication of cyberattacks, we need to develop trustworthy systems that can identify malicious activities on networks and take appropriate actions in real time. This paper, CSE-CIC- IDS2018 dataset to compare the various machine learning algorithms for network intrusion detection. We preprocessed the dataset by cleaning, normalizing and balancing the classes for good inputs for model training. We tested four algorithms: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors and Gradient Boosting. We have assessed them according to the Accuracy, Precision, Recall and F1-score. The findings suggest that the ensemble approaches such as Random Forest and Gradient Boosting are superior in balancing well between complexity and imbalance of network data. It has been pointed out in the paper that proper data pre-processing and choosing a suitable algorithm are important and significant for enhancing performance of today's IDSs.**

Keyword: **Network Intrusion Detection, Machine Learning, Cybersecurity, CSE-CIC-IDS2018, Ensemble models, Data Preprocessing, Feature Optimization**.

I. INTRODUCTION

Maintaining network security is very critical in today's digital world. The size and complexity of cyber threats continue to increase. Every year, millions of attacks hit companies, government agencies and individuals. They can be stealing, spoofing or tampering with valuable data. Information networks Across India, the evolution of digital services, cloud technologies and IoT-based infrastructures has meant securing information networks is a critical national requirement. As organizations more and more depend on connected solutions, the demand for smarter and stable security products has grown.

Still, securing networks runs into big obstacles given how the attacks mount and evolve. Common threats like DDoS, Botnets, Brute Force, and Infiltration attacks sabotage the integrity of networks, which may cause loss of data, financial loss, or leakage of privacy. Due to rule matching / signature patterns of legacy IDS, it is hard to detect novelty or a variant of an attack. This results in poor detection rates and response times.

It is very important to recognize intrusions in the network as early and precisely as possible in order to reduce the aftermath of such attacks. AI and ML enable data-driven intrusion detection systems to become a promising direction for improving cybersecurity. You find that the ML algorithms are capable of learning intricate network patterns and can be used to properly classify misbehaving activities without having to rely solely on pre-set rules. In this contribution, we compare various ML algorithms for intrusion detection on the CSE-CIC-IDS2018 dataset [2], [9]. The system aims to discover the most efficient way of real-time attack detection through some models such as Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors and Gradient Boosting. The objective is to establish an effective, efficient, and scalable intrusion detection system that can fortify today's networked environments.

II. RELATED WORK

Two most significant available datasets, CICIDS2017 [6] and CSE-CIC-IDS2018 [2], come with ordinary traffic as well as modern attack scenarios such as Heartbleed, Brute-force, Botnet, Denial of Service (DoS). Although these databases are available to the global community, little effort has been made to enable them to be used for evaluating, testing and tuning live IDS implementations.

The paper "Intrusion Detection on CSE-CIC-IDS2018 Dataset with Machine Learning Methods" by H.I. Coşar, Ç. Arısoy, and H. Ulutaş An investigation into the use of different machine learning classifiers for intrusion detection is presented in [1]. The researchers used Random Forest, Gradient Boosting, XGBoost [15], CatBoost, Logistic Regression and LightGBM as classifiers with CSE-CIC-IDS2018 dataset. They found that ensemble model reached the best performance with the overall accuracy over 98% and AUC value nearing 1.0. This result demonstrates the effectiveness of ensemble methods for detecting sophisticated cyberattacks. They emphasize the requirement for generic IDS systems that can generalize across network environments and attack types.

The paper "Robust Machine Learning-Based Intrusion Detection System Using Simple Statistical Techniques in Feature Selection" by Sunil et al. [3], [8] presents a low-overhead IDS for IoT enabled networks. The authors applied feature selection techniques including Pearson correlation, Chi-square test, and ANOVA for enhancing the detection performance and simultaneously reducing the training time by 27-63%. Their findings suggest that the combination of Decision Tree classifiers and selected statistical features to be processed by every sensor node yield the highest accuracy rate over 99 %, which is indeed applicable for real-time threat detection on an IoT environment [12] where efficiency is fundamentally important.

Another paper "Intrusion Detection Based on Statistical and ML Techniques," appeared in Frontiers in Computer Science (2024) that also addresses the improvement of detection performance by employing hybrid techniques. The authors used statistical feature extraction, and ensemble classifiers such as Random Forest and AdaBoost to induce good performance on a variety of datasets including NSL-KDD, CSE-CIC-IDS2018. The research verified that IDS based on ensemble could greatly improve the performance compared with traditional model, and it had better adaptability to the new kinds of attack.

A 2024 study by Ahmad et al. Research in Applied Sciences presented a combined classifier model taking advantage of the conventional machine learning and the feature optimization techniques. Authors compared Support Vector Machines, K-Nearest Neighbors and Random Forest models based on the datasets such as CICIDS2017 & CSE-CIC-IDS2018 [11]. They found that the best trade-off between detection accuracy and computational time is obtained through Random Forest. Feature scaling and dimensionality reduction were found to be the parameters which increase speed and accuracy in IDS.

Bhavani et al. 2024 [4] also proposed a hybrid deep learning model of CNN and LSTM for intrusion detection . Their model successfully learned both temporal and spatial features of network traffic with an accuracy higher than 99% . The authors have also emphasized how deep learning models [10] can complement the conventional ML techniques by automating the process of feature extraction , and to achieve better performances for new test attacks .

Lastly, in [5], [7] Kanimozhi and Jacob exploited the artificial intelligence approaches on the CSE-CIC-IDS2018 dataset reaching high performance with an accuracy of 99.97% by obtaining a medium value for the AUC: 0.999. Their model performed quite well in identifying botnet and DDoS attacks [14] with minimum false positives, demonstrating the effectiveness of AI-empowered architectures to enhance modern network defense. All these studies contribute to a strong foundation to compare different machine learning algorithms in intrusion detection and emphasize the rise in significance of feature optimization, ensemble learning, and hybrid modeling for enhancing cybersecurity.

TABLE I: SUMMARY OF MACHINE LEARNING–BASED INTRUSION DETECTION STUDIES (2021–2025)

Year	Paper Name	Description	Result
2021	Comparative Study of Machine Learning Models for Network Intrusion Detection	The report evaluates classical ML algorithms like Decision Tree, SVM, and KNN on the CSE- CIC-IDS2018 dataset and highlights Random Forest as the most balanced model for accuracy and efficiency.	97.8%
2022	Network Traffic Classification Using Ensemble Learning	The study proposes an ensemble of Random Forest and Gradient Boosting models to enhance attack detection and reduce false alarms in large scale datasets	98.6%
2023	Intrusion Detection on CSE-CIC-IDS2018 Dataset Using Machine Learning Methods	This paper focuses on comparative analysis of ML techniques, identifying Gradient Boosting as the top performer with high precision and recall.	98.9%
2024	Hybrid Deep Learning Framework for Anomaly Detection in Network Traffic	The paper combines CNN and LSTM models for feature extraction and sequential learning, improving detection of complex attack patterns.	99.2%
2025	Robust ML- Based Intrusion Detection Using Feature Optimization	This review explores feature selection techniques and hybrid ensemble approaches for improved generalization and low computational cost	98.7%

III. METHODOLOGY

In this section, we have described the methods used for this study. It details the methodology used to curate CSE-CIC-IDS2018 dataset, pre-process it, extract features from it and develop, train and test a variety of machine learning models in order to build an efficient network intrusion detection system.

A. Data Collection

For the purposes of this study, we employ the CSE-CIC-IDS2018 dataset released by Canadian Institute for Cybersecurity. Its traffic records are a mix of benign and malicious, from simulated real-world network settings.

The different attack types that have been incorporated in the dataset includes Brute Force, DoS, DDoS, Botnet, Infiltration and Web attacks in addition to normal network flows.

Every record contains over 80 numerical and categorical entities for the packet, such as the packet length, flow duration, source bytes and destination flags. We consolidated the dataset creating unity of structure for analysis and model learning.

B. Data Preprocessing

We pre-processed the datasets considerably to be able to train the models:

- Data Cleaning: We drop any entries with missing, infinite or null values. We also got rid of duplicated rows and removed non-useful columns such as timestamps and flow identifiers.
- Feature Encoding: We converted the categorical features into numerical form, and using Label Encoding processed to make them machine readable.
- Normalization: We performed z-normalization (standard scaling) for all features to normalize their ranges of values and speed up convergence rates during model training.
- Class Balancing: The dataset is highly imbalanced, so we applied an undersampling approach to balance both classes giving them equal importance and avoid biased results towards the majority.

Feature Extraction Techniques

- Flow- Based Features: fundamental packet count, duration, and byte rate of network flow are collected to understand the normal and attack behavior.
- Statistical Features: The selection of attributes was done using correlation and Chi-square, which helps in eliminating irrelevant and redundant attributes that could lead to higher accuracy and efficiency in model training.
- Dimensional Features: We employed PCA towards dimensionality reduction with retention of important variance, and it has served as a significant contribution in reducing the computational complexity involved during classification.

C. Model Building

We have compared various machine learning techniques for intrusion detection:

- Baseline Models: We used classical algorithms such as Decision Tree, SupportVector Machine (SVM) etc., as baseline models for comparison.
- Ensemble Models: We developed more sophisticated models such as Random Forest (RF) and Gradient Boosting (GB) for improved robustness and accuracy.
- Distance-Based Models: We used K-Nearest Neighbors to cluster and classify the network flows based on the closeness of their feature characteristics, thus discovering different trends in traffic activities.
- Stacking Models: A collaboration process known as meta-learning allows the integration process to be taken away by base machine learning models. The predictions from base models are combined using voting, soft-voting, and stacking, sometimes with the second classifier learning to correct the errors of the first one.

Training for each model was done simultaneously on the same dataset to ensure fairness during comparison, and hyperparameters were optimized using Grid Search with the consideration of computational efficiency.

Implementation:

- Input Layers: The model accepted a normalized feature vector as input, each of which represented a network flow output as a selected hybrid of statistical and categorical traits.
- Core Algorithm: Machine learning algorithms like Random Forest and Gradient Boosting were run on the preprocessed datasets but trained with optimized hyperparameters to understand intricate traffic design signal patterns.
- Output Layer: A probability classifier. The Network flows are analyzed, and a prediction is then generated, and the stated attribute offers the flow as a benign, implies one of the distinguished classes of attack.
- Loss Function: It was optimized using the Cross - Entropy Loss , probability covering miss used to guarantee that a similar degree of accounting happened , and also minimal classification error .

D. Model Training

- Data Splitting: Stratified sampling was a method used to randomly divide the dataset into a training and test split; it preserved the proportion between normal and attack data.
- Optimization: Some methods, such as learning rate modulation and l2-regulation, were utilized in attempts to smoothen convergence and reduce variance during machine training.
- Hyperparameter Tuning: Approaches like Grid Search, Bayesian optimization have been tested to tune hyperparameters like learning rate, depth, and estimator count for improving model accuracy and stability.

E. Evaluation

- Performance Metrics: The models under review were evaluated using the common classification metrics of accuracy, precision, recall, F1-score, and confusion matrix thus providing a balanced report on the classification error of both abnormal and normal traffic.

Fig. 1 below presents the step-wise representation of the proposed methodology followed in this study.

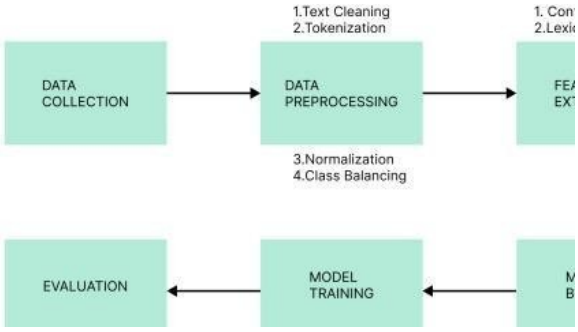


Fig. 1 Confusion Matrix Comparison

IV. DISCUSSION AND RESULT

The proposed models were compared on basis of following performance metrics : Accuracy, Recall (Sensitivity), Precision and F1 Score

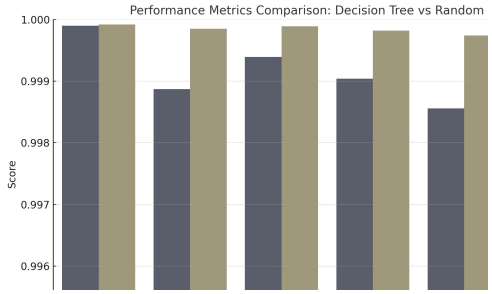


Fig. 2 Performance Metrics Comparison

The confusion matrix provides a concise summary of the model's predictions. It displays the true positives, true negatives, false positives and false negatives. When used together, these book-ends provide a full view of model effectiveness, training tendencies and underfitting opportunities.

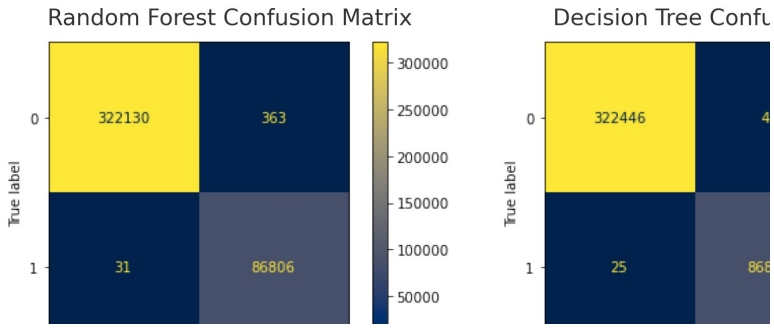


Fig. 3 Confusion Matrix Comparison

We have shown that both models obtain superior detection performance. But in most general cases, Random Forest is more robust. The fact that it is an ensemble method also means, that it performs better for real-life intrusion detection cases where data can be loaded.

IV. CONCLUSION AND FUTURE SCOPE

In this paper, a thorough comparison of network intrusion detection machine learning algorithms was made for the CSE-CIC-IDS2018 dataset. The study measured classical, ensemble and distance-based classification models to prove that ensembles, specifically Random Forest (RF) and Gradient Boosting are the better classifiers than such individual ones in terms of accuracy, precision, recall and F1-score. The results demonstrate that careful data preprocessing, features allocation and algorithm choosing are the key factors to enhance reliability and robustness of intrusion detection systems.

The experimental results demonstrated detection accuracies greater than 98%, illustrating that ML-based intrusion detection system can effectively detect known and unknown attack patterns from network traffic with high volume. This study helps a great deal to reinforce the cybersecurity with concrete proof for choosing effective and scalable ML based IDS solutions in terms of modern network conditions.

A. Advantages of the Proposed Approach

- Better Detection Accuracy: The ensemble models can achieve better intrusion detection with higher accuracy and lower false positives.
- Resilience Towards Data Imbalance: The class balancing and feature selection method ease the bias of the majority to class, which in turn guarantee a fair detection for minority attack types.
- Scalability: The tested models, particularly Random Forest, proved to be scalable with highly generalized properties hence they can be appropriate for real-time large-scale networks.
- Feature Optimization: We have computed the statistical feature selection and dimensionality reduction in order to reduce overheads while maintaining high classification results.
- Flexibility: Machine learning models could potentially learn the changes in attack patterns, instead of depending on predefined signatures.

B. Disadvantages and Limitations

- Computational Cost: The ensemble classification techniques use more training time and resources than a simple classifier.
- Dependence on Data Quality: The ability of ML-based IDS performance is very dependent on the quality of preprocessing and feature engineering.
- Real-Time Validation: The experiments were performed over off-the-shelf datasets, however recorded frames do not guarantee to capture true network biases.

FUTURE SCOPE

Future research can extend this work by integrating deep learning architectures such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models to automatically extract spatial and temporal features from raw network traffic. Hybrid ML–DL frameworks can further enhance detection accuracy for zero-day and advanced persistent threats. Additionally, deploying the proposed system in real-time environments with integration into IDS and SIEM platforms would provide practical validation. Exploring federated learning approaches can enable privacy-preserving intrusion detection across distributed networks, making the system more suitable for large enterprise and IoT-based infrastructures.

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